

DISMANTLING THE STIGMATIZATION IN “ENVIRONMENTAL ESTROGEN”

By Group UNDISIM

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Members : Wan Peng Zhuanghuan Yin Qiyun Hu Anqi Liu



A goal of our
presentation

1. Review and
social phenomenon
statement

2. Solid evidence of
the problems in
toxicological
researches

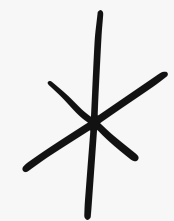


3. people's
misunderstanding
of estrogen and
solution

4. Estrogen and
mental "one
health"

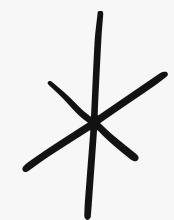
PREVIOUSLY ON OUR TOPIC

Our mid-term presentation



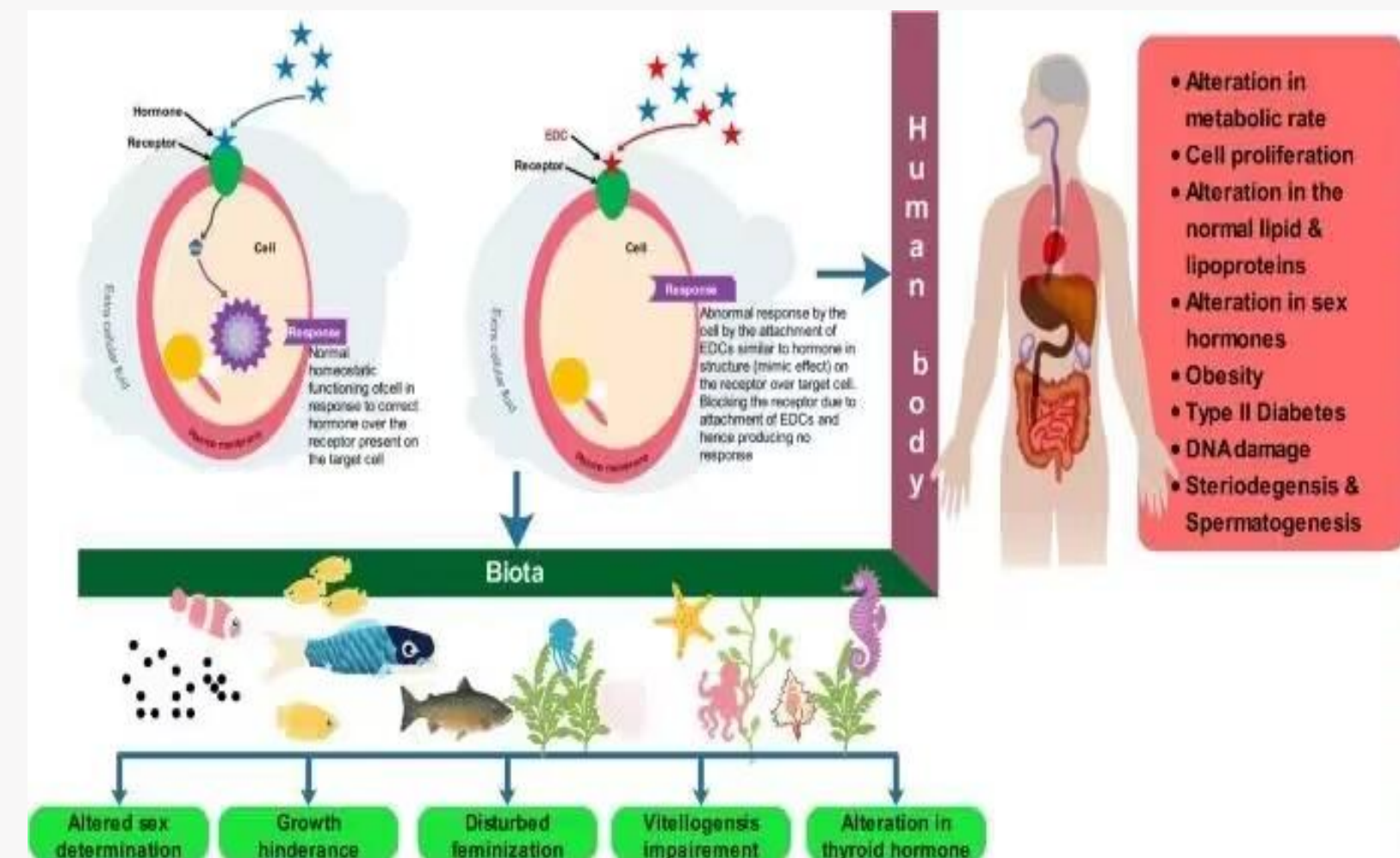
What we've told

We've already talked about basic information about estrogen, including its definition, its transfer mechanism, source and impact.



Some doubts

At the end of our last presentation, we've raised some questions about how people treat "feminization", how people treat scientific research rationally



The impact and transfer mechanism of EDC

SOCIAL PHENOMENON

Result from the problem of researches



Afraid of feminization



People begin to panic on eating food contain
estrogen such as soybean milk

OUR POPULAR SCIENCE VIDEO ON ESTROGEN

01

02

03

04

05

06

Part 1

What is **EDC**

Part 2

Source of it

Part 3

types

Part 4

effects on human
health

Pictures

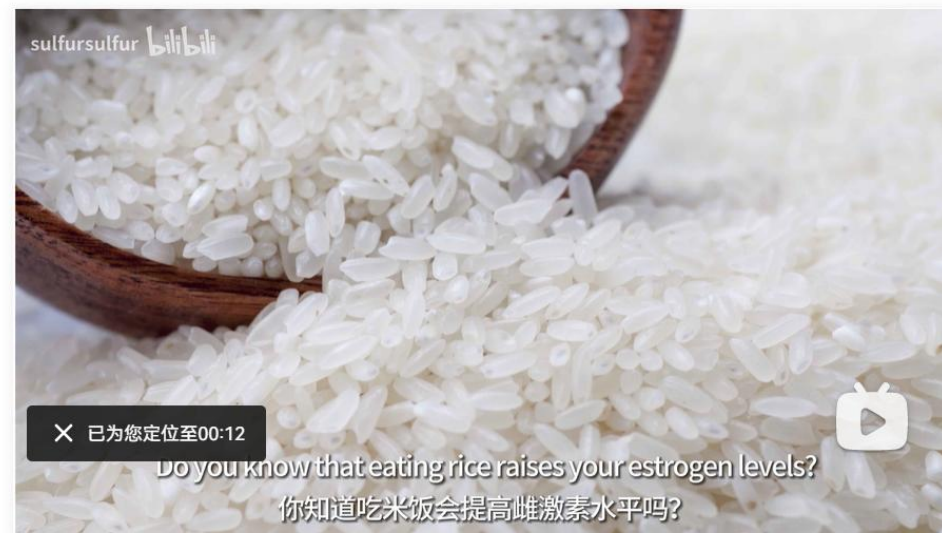
From our mid-term
text or from articles

Attractive points

At the beginning of
the video, we add
some terrifying
facts and gave it a
title “terrifying
environmental
estrogen”.

【全英科普视频】the terrifying environmental estrogen

137 0 2023-05-15 16:34:14 未经授权，禁止转载



1 人正在看，已装填 0 条弹幕

发个友善的弹幕见证当下

弹幕礼仪 > 发送

THE VIDEO MEETING

05
Review

Look back on our topic, prove what we've think and find a clue to state the relationship between estrogen and mental one health(making our logic clearer)



04
Online meeting

Discussed with Dr. Packer about the problem of present research have and the detail of the solution as well as estrogen and mental health



03
think

Rethink about the goal of the topic and give out more uestions and views

02
Ask

e-mail Dr. Packer for an answer

01
Discuss

Read through the article carefully and raise question

SPECIAL SECTION

What's Gender Got to Do with It? Dismantling the Human Hierarchies in Evolutionary Biology and Environmental Toxicology for Scientific and Social Progress*

Melina Packer^{1,†} and Max R. Lambert²

1. Department of Gender and Sexuality Studies, University of California, Riverside, California 92521; 2. Department of Environmental Science, Policy, and Management, University of California, Berkeley, California 94720

Submitted June 17, 2021; Accepted January 18, 2022; Electronically published May 10, 2022



Article is here!

A questionnaire on estrogen and gender equality



CHECKLIST

- ☒ Released a questionnaire
- ☒ Hold activities on campus
- ☒ Online meeting with Dr. Packer
- ☒ Making a popular science video on estrogen
- ☐ Final analyzation

SOCIAL PHENOMENON

Result from the problem of researches



Afraid of feminization
/food contains estrogen



People begin to panic on eating food contain
estrogen such as soybean milk

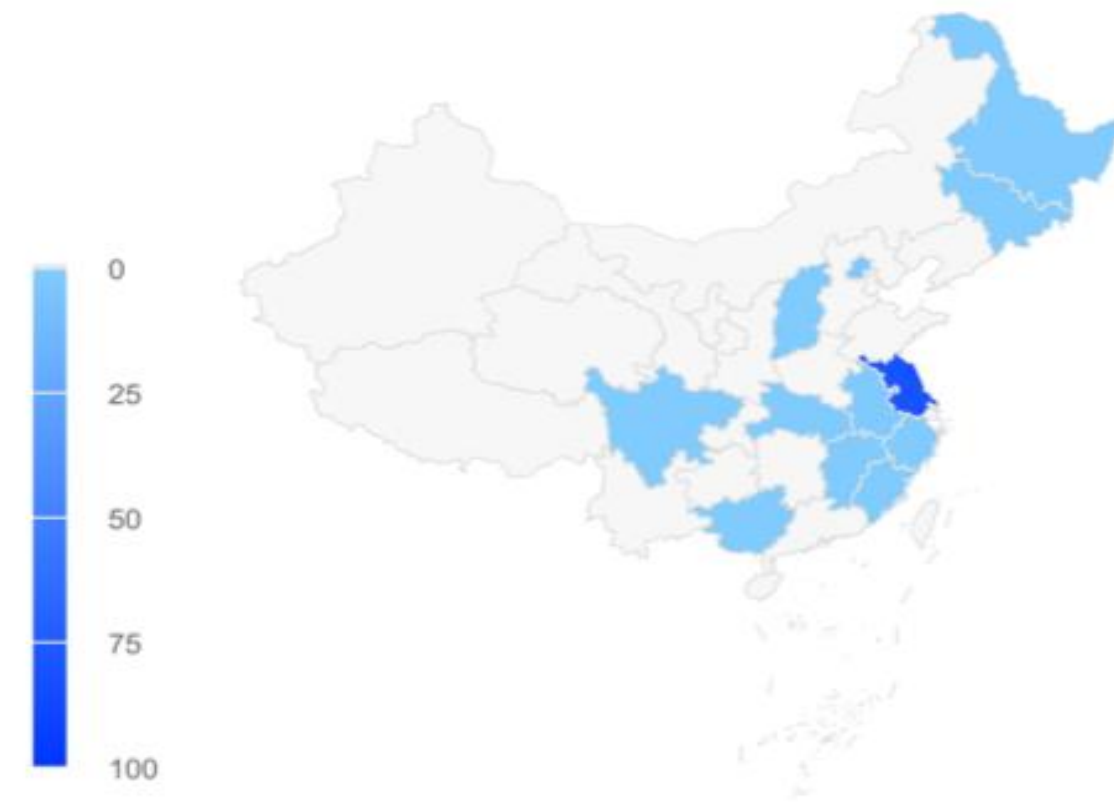


OUR CURRENT PROGRESS

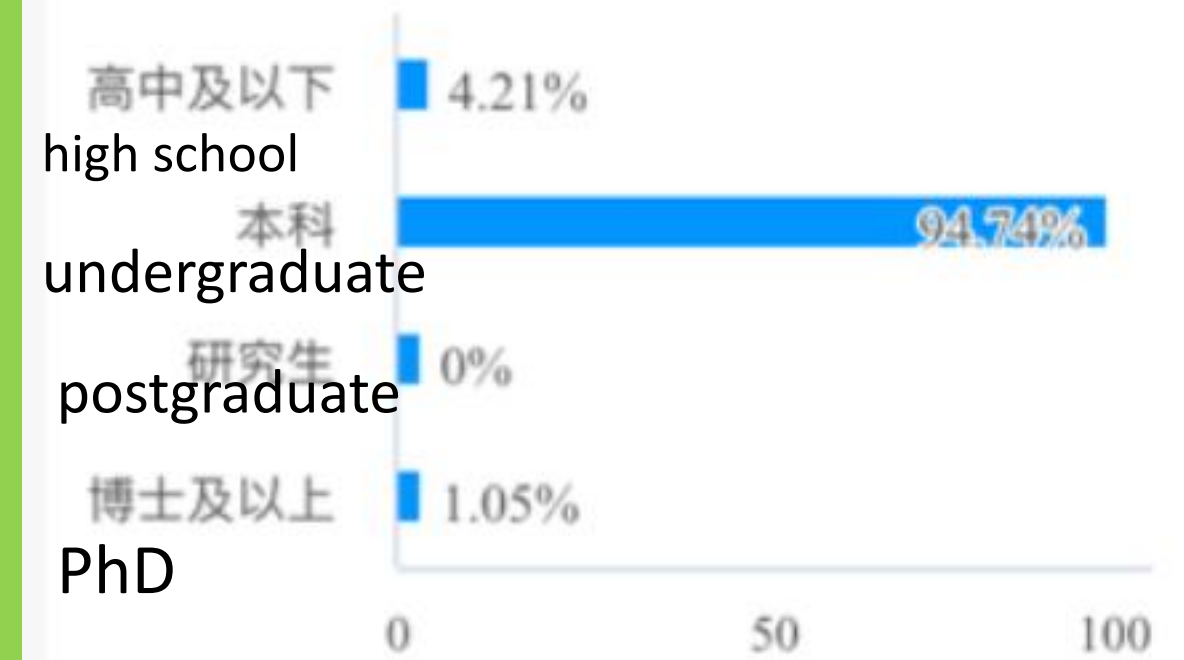
1.性别 Gender



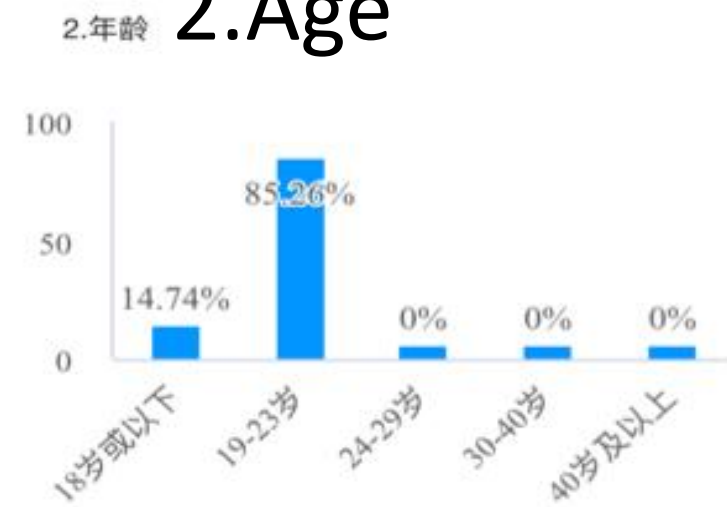
位置分布 Location



3.学历 3.Degree



2.年龄 2.Age



Fundamental Identification

5.是否了解环境雌激素这一概念

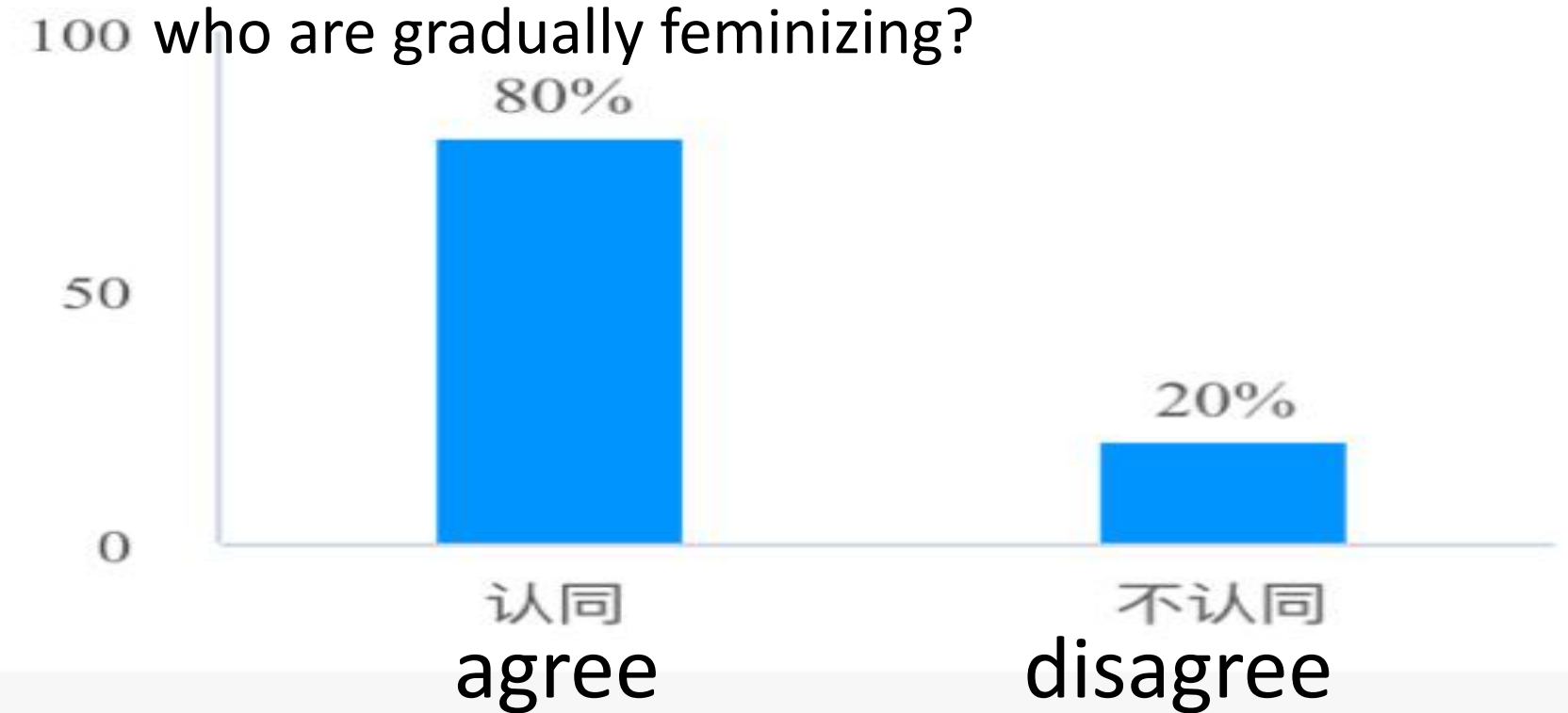
Weather familiar with the environmental estrogen



Detail information

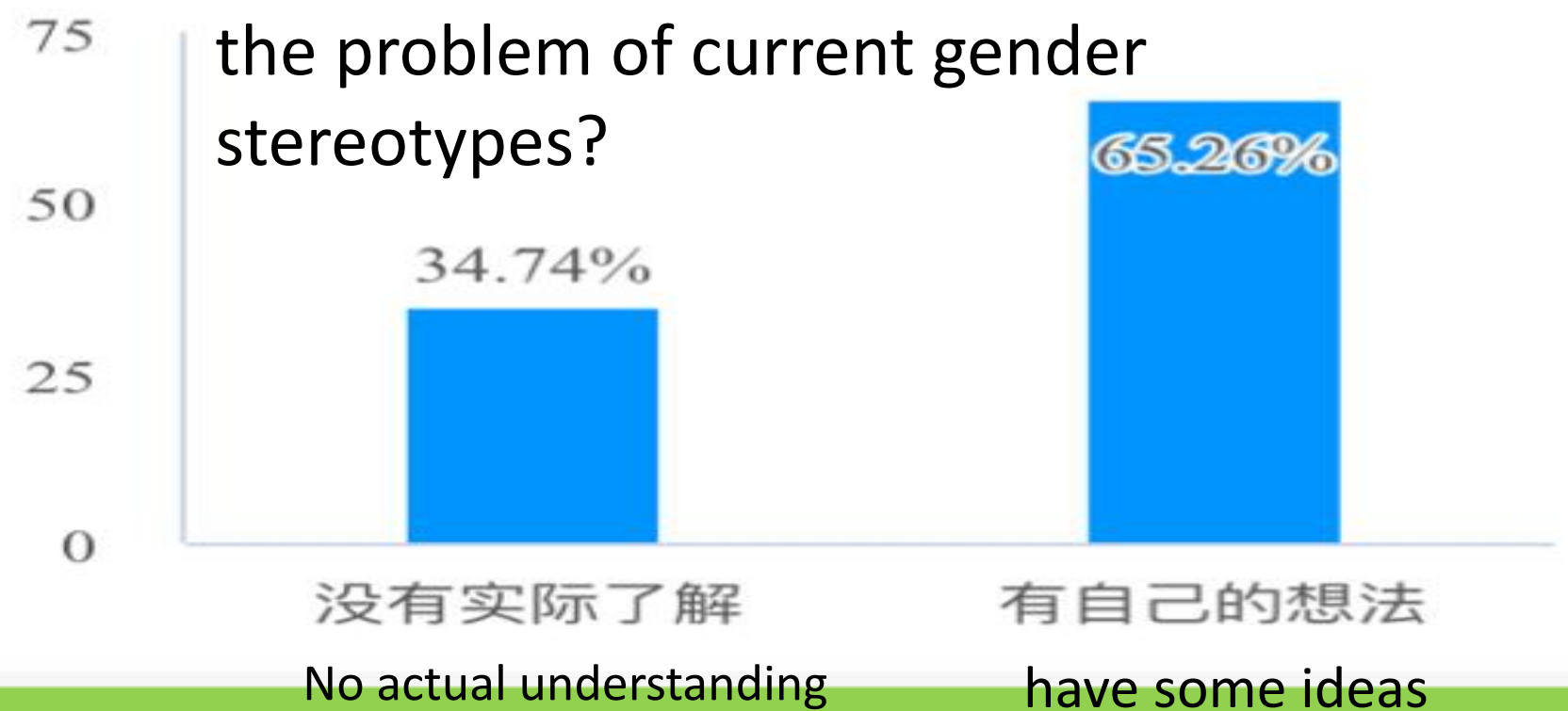
8.你是否认为当下有一批男性逐渐女性化

Do you think there is a group of men who are gradually feminizing?



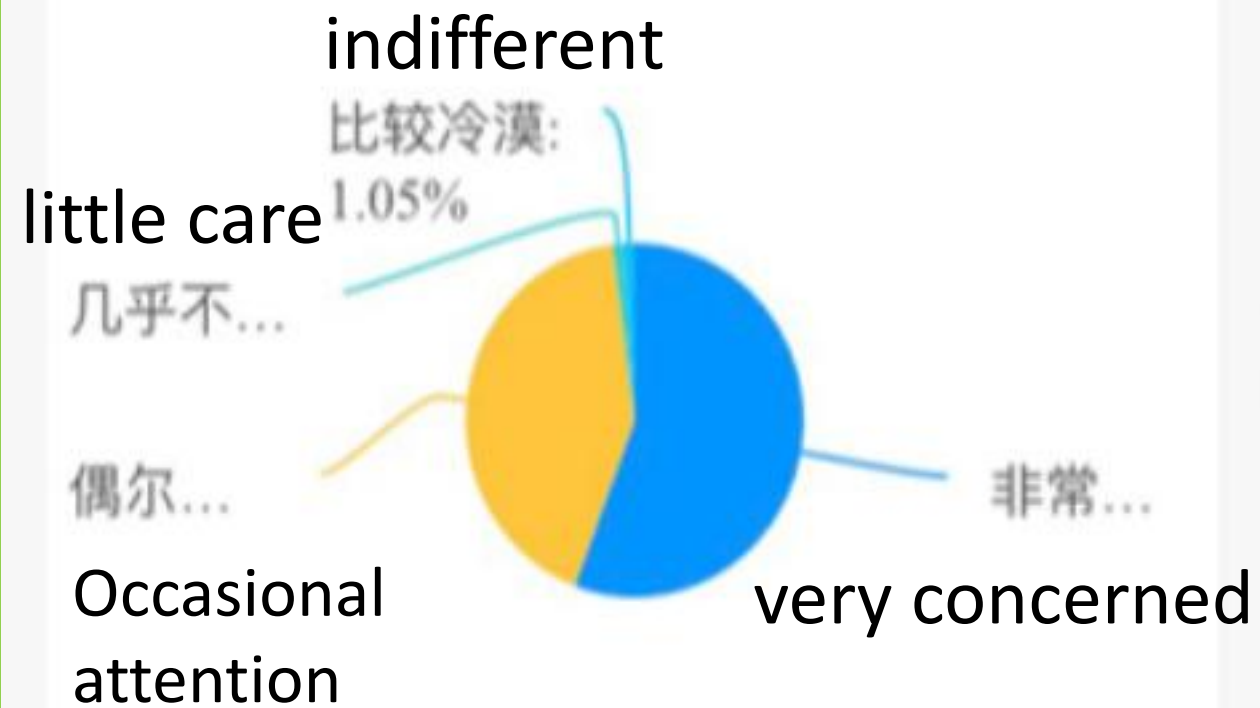
11.请问您是否有从当下的刻板印象认识出亟...

Have you thought of solutions to the problem of current gender stereotypes?



4.对当下性别不平等的问题的看法

Views on current gender inequality



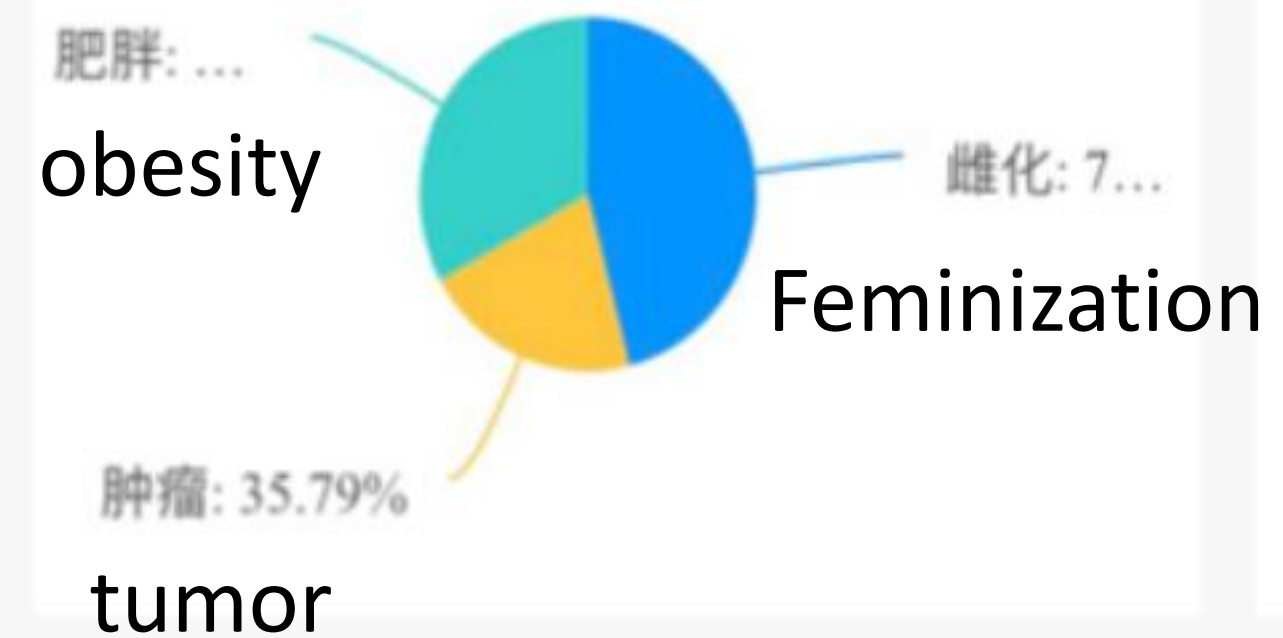
6.大家对于性激素的了解来源

Source of understanding about estrogen



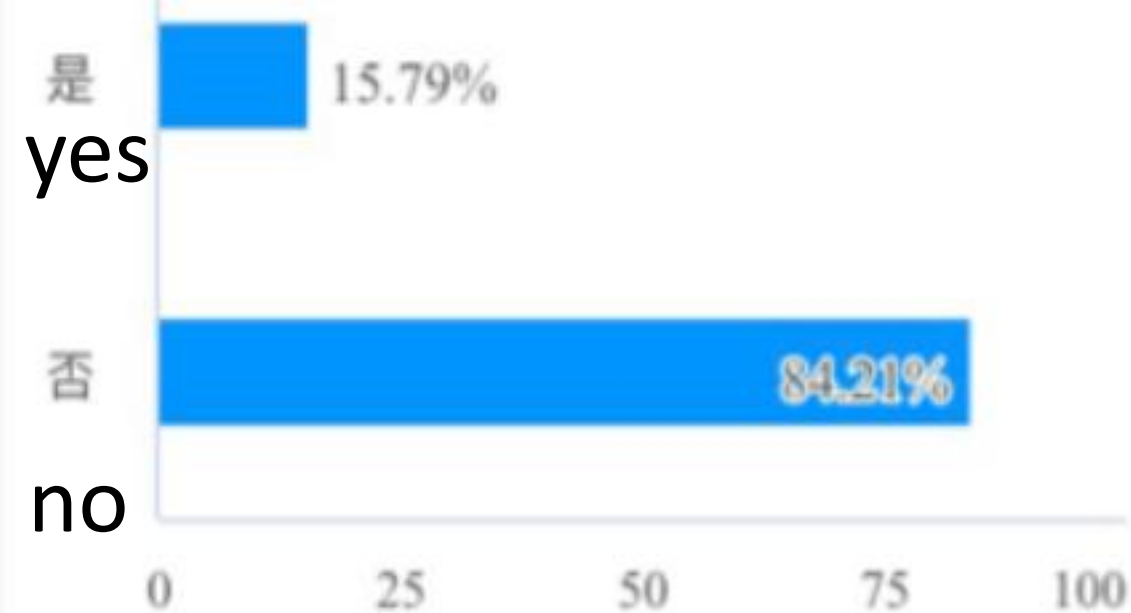
7.您认为如果体内雌激素过高,会有怎样的...

Effects of high levels of estrogen in the body



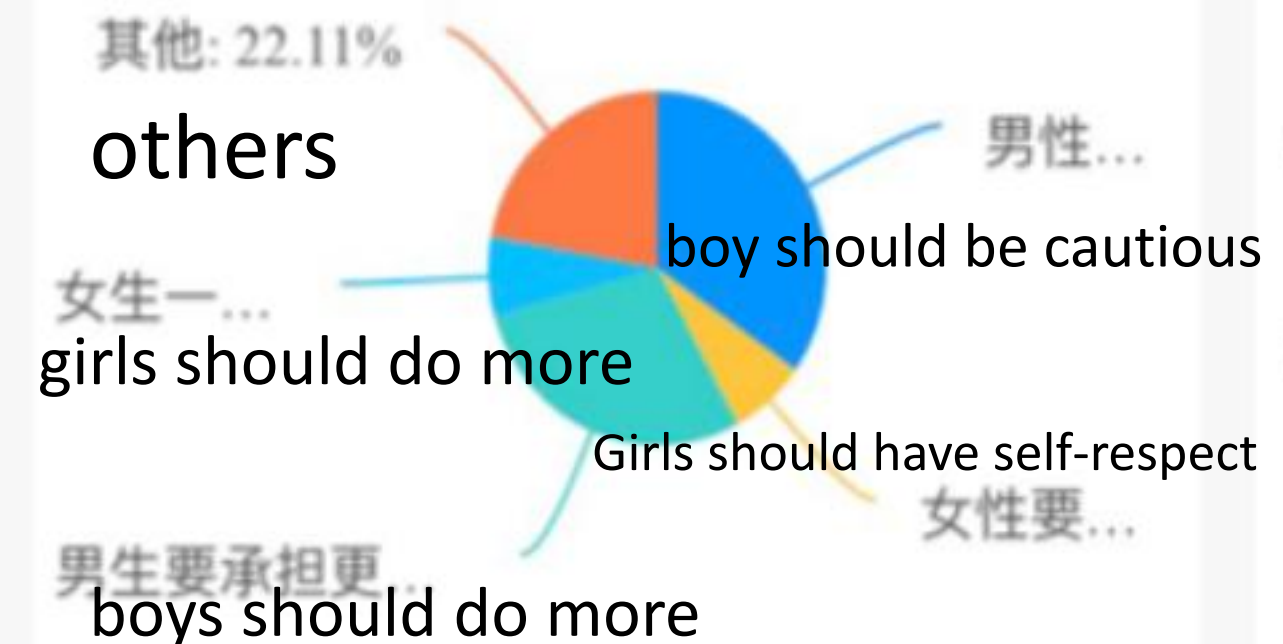
9.请问您认为男性雌性化这一现象就是雌激...

Do you think the reason for male feminization is estrogen?



10.你觉得在你所了解的有关环境雌激素的报...

Viewpoints on the relevant reports that have been understood



key words

Long live thought/ humanity/society
/Structuralism/be yourself/Chinese education/
influence/gender/
stereotypes/concepts/history/less rigid/internet
inclusivity/respect/equality/division of
labor/contradictions

our message board!





interesting
meaningful

positive
warm
confident



Problems of toxicological research

* Many studies are focused on the discussion of reproductive behavior, sexual organs and gonads

* Biologists overwhelmingly assume that “abnormal” reproductive physiologies both are generated by EDC exposure

[Exposure to modern, widespread environmental endocrine disrupting chemicals and their effect on the reproductive potential of women: an overview of current epidemiological evidence.](#)

Karwacka A, Zamkowska D, Radwan M, Jurewicz J.

Hum Fertil (Camb). 2019 Apr;22(1):2-25. doi: 10.1080/14647273.2017.1358828. Epub 2017 Jul 31.

PMID: 28758506 Review.

[Endocrine disrupting chemicals and breast cancer: a systematic review of epidemiological studies.](#)

Wan MLY, Co VA, El-Nezami H.

Crit Rev Food Sci Nutr. 2022;62(24):6549-6576. doi: 10.1080/10408398.2021.1903382. Epub 2021 Apr 5.

PMID: 33819127

[A systematic review: Impact of endocrine disrupting chemicals exposure on fecundity as measured by time to pregnancy.](#)

Kim YR, Pacella RE, Harden FA, White N, Toms LL.

Environ Res. 2019 Apr;171:119-133. doi: 10.1016/j.envres.2018.12.065. Epub 2018 Dec 28.

PMID: 30660918

[Female exposure to endocrine disrupting chemicals and fecundity: a review.](#)

Mínguez-Alarcón L, Gaskins AJ.

Curr Opin Obstet Gynecol. 2017 Aug;29(4):202-211. doi: 10.1097/GCO.0000000000000373.

PMID: 28557831 [Free PMC article.](#) Review.

[Neurodevelopmental impact of the offspring by thyroid hormone system-disrupting environmental chemicals during pregnancy.](#)

Salazar P, Villaseca P, Cisternas P, Inestrosa NC.

Environ Res. 2021 Sep;200:111345. doi: 10.1016/j.envres.2021.111345. Epub 2021 Jun 1.

PMID: 34087190 Review.

Problems of toxicological research

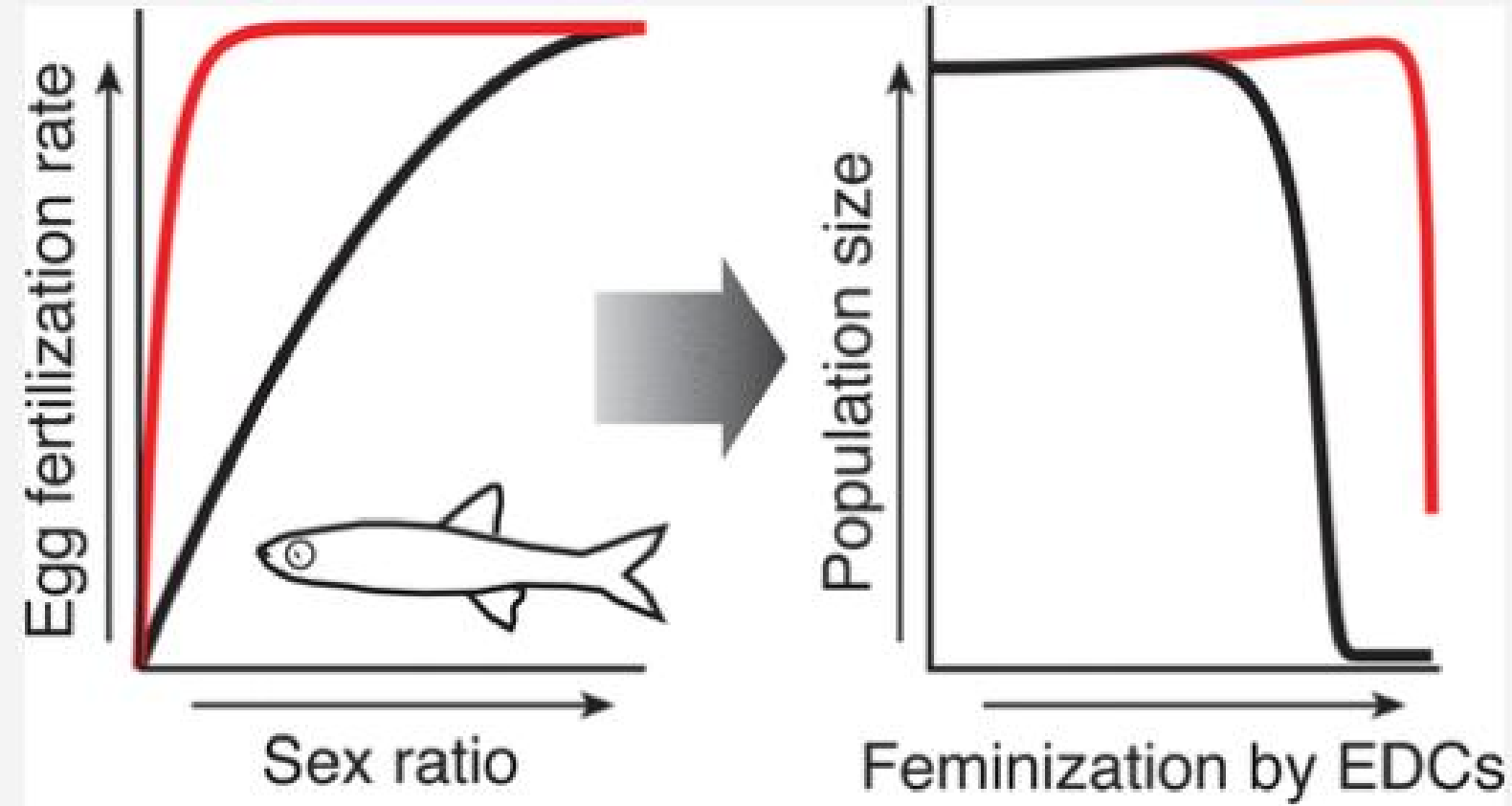
* Toxicology often focuses on end points that putatively correlate with biological fitness.

These assumptions are overwhelmingly extrapolative : not empirically evidenced from wild populations.



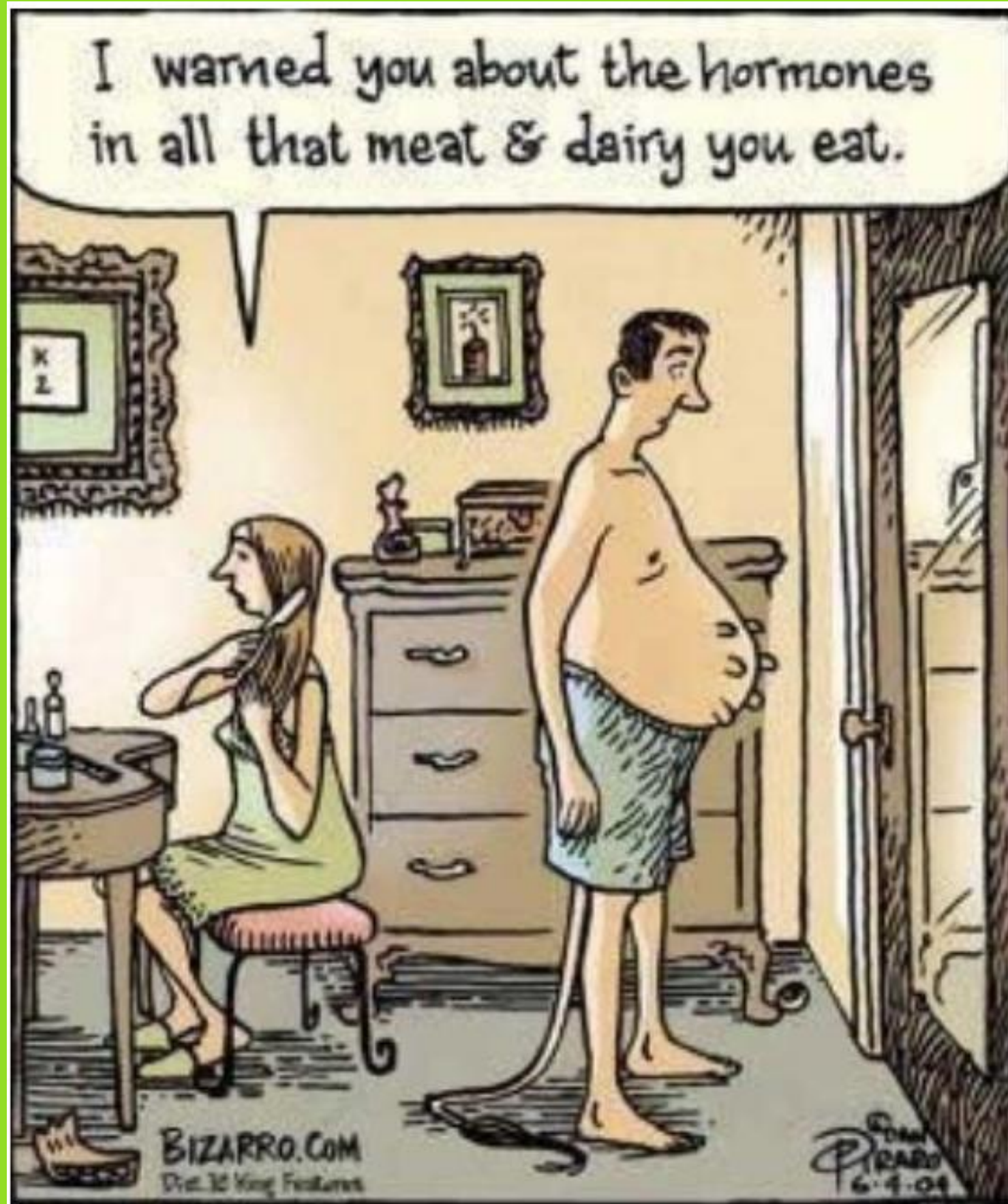
limits toxicology's contributions to fundamental biological research while also distracting from pressing environmental justice needs

- ◆ The effect of estrogen on the population also depends on the male : female ratio of the population and other related factors



For Example

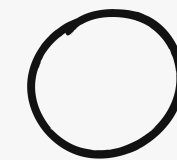
- A study on menidia beryllina : effect of EDC on the fish was largely dependent on the status of mating function
- A seven year microsatellite field study : fish can overcome the effects of EE2



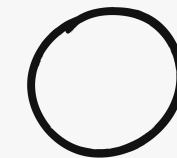
Formation of prejudice



The limitation of toxicology research



"Unjust" outcome in society



Our wrong concerns

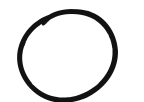
Disconnected from pluralistic society

01

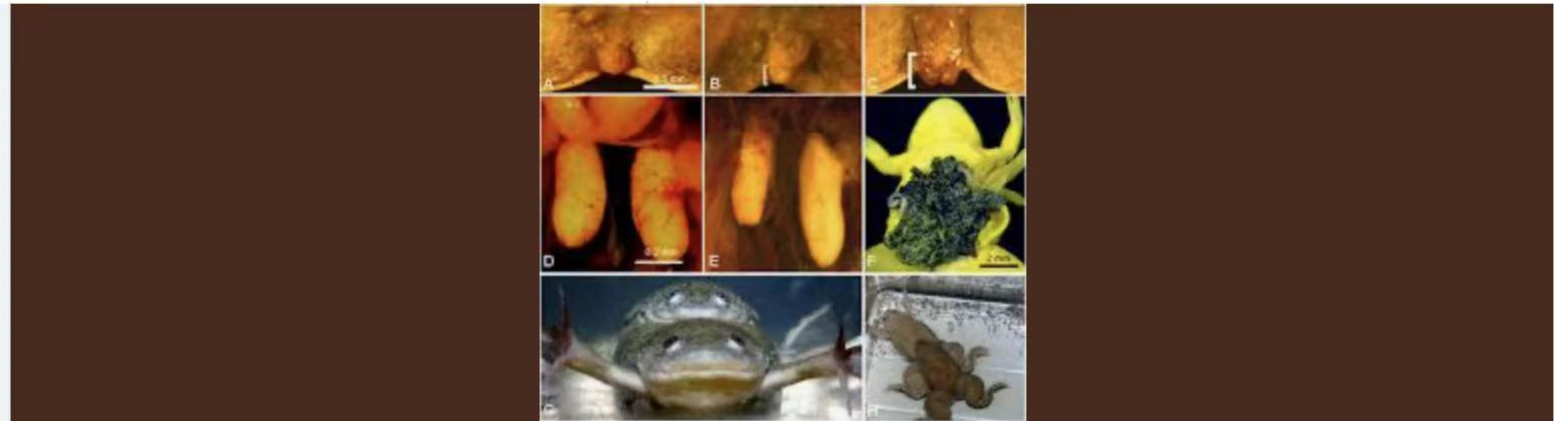
- Gender, sexual orientation ability...
- Dominant groups
- **Natural** selection



“normal” “natural”



“abnormal” “polluted”



Atrazine-exposed males suffered from depressed testosterone, decreased breeding gland size, demasculinized/feminized laryngeal development, suppressed mating behavior, reduced spermatogenesis, and decreased fertility. These data are consistent with effects of atrazine observed in other vertebrate classes. 2010年3月9日



More scientists!!!

What it brings us?

02 Such EEs leads to public anxieties

* “Chemical castration” (化学阉割)

Transmisogyny
(跨性别歧视)

discrimination

* Anxiety:

feminization of populations ~~~~~ male extinction

diversification of sexual variation ~~~~~ extinction of gender normativity



Example 1: A farce of “mermaid”

“Detergent pesticides
spoil my sex drive”

“Give me back my masculinity”

Fish: “abnormalities” and “deviation”
from their “natural” sexuality

Dangerous

- Biological reproductive disorders
- Stunt child growth



Vegan men are more feminine than omnivorous men

Vegetarian men

Soybeans

Meat

- Plant-based diet
- Inclusive and flexible masculinity

SOYBEANS

Stigmatizing estrogen

Estrogen: Isoflavone

Health diet
Less greenhouse gas



Other dangers?

=



Modern trends

Soybean isoflavone exposure does not have feminizing effects on men: a critical examination of the clinical evidence

Mark Messina Ph.D.  

- **cancer**
- **diabetes**
- **internal organ disorders**

Some previously published research can be downright scary, with claims that increased soy can mess with your hormones, the thyroid, and possibly cause cancer. [D]oes soy deserve that health halo, or should you swear the stuff off of your shopping list for good?

—Mary Grace Taylor and Kate Rockwood, *Good Housekeeping*, 2020



make advocations



ADVOCATE



set policies



As message receivers:



- ① cautiously
- ② critical thinking
- ③ don't swim with the stream

BUT!

Multidisciplinary Collaboration

diversity of human society



diversity of species



diversity of the environment



more inclusive and open mind



Mental One Health

**THANK YOU
FOR YOUR
TIME!**

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